

5.4.3 Apply Mitigation Techniques

تحفظ کی تدابیر خطرات کو روکتی یا ان کا نقصان کم کرتی ہیں: اینٹی وائرس اور فائر وال، مضبوط پاس ورڈ اور دو عنصری...

CS XII

Chapter 5: Application and Impacts of Computing

Lecture 12 of 15

5.4.3 Apply Mitigation Techniques

LEARN IT

Mitigation techniques are actions taken to **prevent** security threats or **reduce their harmful effects** once identified. Technology helps, but it cannot stop all threats if the user is careless.

#	Technique	How it helps
1	Antivirus and firewalls	Antivirus scans and removes harmful software (for example from USB drives). A firewall monitors network traffic and blocks suspicious access from the internet.
2	Strong passwords and 2FA	Complex passwords slow brute-force guessing. 2FA adds an extra layer: an OTP on your phone as well as the password.
3	Data encryption	Converts data into unreadable code during transmission. Protects passwords and sensitive data if they are intercepted.
4	Regular backups	Copies on cloud or external drives allow quick recovery after a crash, human error or ransomware.

5	User awareness (the human factor)	Training users to avoid unknown links, suspicious attachments and password sharing. The human is often the weakest link — and the strongest defence when trained.
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WORKED EXAMPLE

Worked example: match each threat to a mitigation — BISE e-test week at a college in Larkana.

Threat seen	Mitigation to apply
USB games in the lab, pop-ups starting.	Scan with antivirus ; block unknown USBs; enable the lab firewall .
Phishing SMS about “update your portal password.”	User awareness : do not click; verify on the official site typed by hand. Turn on 2FA .
Fear that intercepted Wi-Fi could show passwords.	Use sites with encryption (HTTPS / encrypted LMS); never send passwords in plain WhatsApp.
Ransomware risk to project files.	Regular 3-2-1 backups so files can be restored without paying.
Shared passwords on a sticky note on the monitor.	Strong unique passwords + 2FA; remove the note; awareness session.

REMEMBER IT

Five shields: A-F, P-2, E, B, U. Antivirus+Firewall, Passwords+2FA, Encryption, Backups, User awareness. Ransomware is why backups matter: if files are locked, a clean copy brings them back.

COMMON MISTAKE

Mistake: “If I installed antivirus I am 100% safe.”

Fix: Antivirus does not stop you from typing an OTP into a fake page. Mitigation is a **set** of techniques plus human caution.

تحفظ کی تدابیر خطرات کو روکتی یا ان کا نقصان کم کرتی ہیں: اینٹی وائرس اور فائر وال، مضبوط پاس ورڈ اور دو عنصری تصدیق (2FA)، انکریپشن، باقاعدہ بیک اپ، اور صارف کی آگاہی۔ ٹیکنالوجی تب ناکام ہو جاتی ہے جب صارف نامعلوم لنک کھولے یا پاس ورڈ بانٹے۔

CHECK YOURSELF

1. How does 2FA add a layer on top of a password?
2. Why do regular backups help after ransomware?
3. What three careless behaviours should awareness training stop?

Answers: 1. It also requires an OTP sent to your phone (a second factor). 2. You can restore a clean copy instead of losing the files (or paying). 3. Unknown links, suspicious attachments, and password sharing.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Mitigation means:

- a. Prevent or reduce harm from threats
- b. Ignore them
- c. Rename the threat
- d. Draw them in Figma

MCQ 2. Updates/patches:

- a. Close known holes
- b. Always break Python
- c. Replace HCI
- d. Replace ER

MCQ 3. Least privilege means:

- a. Only the rights you need
- b. Everyone is admin
- c. No passwords
- d. No backups

MCQ 4. User education fights:

- a. Phishing / bad USB / weak passwords
- b. Cosmic rays only
- c. NAND delay
- d. Median bias only

MCQ 5. Incident plan includes:

- a. Isolate, preserve, restore, report
- b. Panic format twice
- c. Pay first always
- d. Post OTPs

MCQ 6. Technical + human controls together are:

- a. Stronger than either alone
- b. Illegal
- c. Useless
- d. Only for banks

Extra short questions

1. Five mitigation techniques from the book.
2. Map one technique to phishing, one to ransomware, one to DoS.
3. What not to do?
4. Why backups must be tested?

Extra long question

1. Write a one-page lab policy: 8 rules that mitigate the threats in 5.4.1. Tag each rule with the threat.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Updates, AV, firewall, backup, MFA, training, least privilege, encryption — any five.
2. Phishing: training+MFA. Ransomware: backup. DoS: filtering/capacity (high level).
3. Share the malware “to warn friends” as an attachment.
4. A backup you cannot restore is theatre.

Long

1. No shared admin (malware/theft); USB scan; official links (phishing); weekly backup (ransomware); firewall; MFA email; report SMS; no pirated cracks. Table.

Next lecture: 5.5 Equity and Equal Access in Digital Collaboration